Comparative Simulation Study of P0, P1, and P2 Hybrid Electric Vehicle Architectures Using MATLAB/Simulink

1) Introduction

Road transportation is a major contributor to global emissions where increasingly stringent emissions legislation and fuel economy targets have driven the development of electrified powertrains as a means of improving efficiency. While battery electric vehicles offer zero tailpipe emissions, their predicted increase in market share remains limited by their lower energy density and higher charging infrastructure and cost. As a result, hybrid electric vehicles (HEVs) present a practical solution, combining an internal combustion engine (ICE) with an electric motor to improve overall efficiency without full reliance on electric propulsion.

One limitation of conventional ICE vehicles arises from their inability to operate consistently within optimal regions of the engine map. Hybridisation addresses this limitation by introducing an additional energy conversion pathway. The electric motor enables regenerative braking, torque assistance and load shifting, allowing the engine to operate closer to high-efficiency regions while reducing fuel consumption and emissions. However, the extent to these benefits is fundamentally set by the architecture of the hybrid system.

Hybrid architectures are typically classified using the P0–P4 framework, which defines the position of the electric motor relative to the engine and drivetrain where this study focuses on the P0, P1 and P2 configurations. In the P0 configuration, the electric motor is connected to the engine via a belt drive, providing limited torque support and restricted regenerative capability. The P1 configuration places the electric motor directly on the engine crankshaft, increasing torque assist but maintaining a fixed mechanical coupling between the engine and drivetrain. In contrast, the P2 architecture positions the electric motor between the engine and transmission and typically incorporates a disconnect clutch, enabling engine decoupling, electric-only operation and more effective energy recovery.

The objective of this study is to compare the performance of P0, P1 and P2 hybrid architectures using MATLAB/Simulink under a Worldwide Harmonized Light Vehicles Test Procedure (WLTP) Class 3 drive cycle to identify performance and emissions benefits based on architecture alone. While previous studies establish general performance trends between architectures, this study focuses on isolating the effect of motor placement under identical control and conditions to better understand architecture driven differences in behaviour, energy usage and emissions.

2) Literature Review

Road transport decarbonisation has increased the need for propulsion systems with reduced fuel consumption and tailpipe emissions without the full cost, mass, and infrastructure dependence required with battery electric vehicles. HEVs have emerged as a result as they keep the conventional ICE while also adding limited electrification to improve efficiency through stop–start operation, regenerative braking, and torque assist. In particular, 48 V systems have gained relevance because they offer a good cost-to-performance ratio, avoid the complexity of high-voltage systems, and can be integrated into existing vehicle platforms much more easily [1].

A HEV combines at least two onboard energy conversion paths, typically an ICE and an electric motor coupled to an energy storage system. At the highest level, hybrid powertrains are commonly grouped into series, parallel, and series–parallel or power-split configurations. Series hybrids decouple the engine from the wheels and use the engine primarily as a generator, whereas parallel hybrids allow both the engine and electric motor to contribute traction torque directly. Series–parallel arrangements combine both energy paths to improve flexibility across a wider range of operating conditions. Although these architectures differ in mechanical layout, their common objective is to operate the engine and electrical system in a coordinated way that reduces fuel consumption and emissions while maintaining drivability [2]. Energy management is therefore crucial to hybrid performance, as it determines when propulsion is provided by the engine, the electric motor, or both, and how battery SOC is maintained during transient driving [3].

For this study, the most relevant subgroup is the parallel mild hybrid architectures, especially the P0, P1 and P2 arrangements. In the P-classification system, the defining variable is the position of the electric motor along the driveline. The P0 architecture places a belt-integrated starter-generator on the accessory belt at the front of the engine. This gives the lowest integration cost and requires the least change to an existing vehicle platform, but torque transfer is limited by the belt drive and energy recuperation is reduced by belt losses and slip. The P1 architecture moves the electric motor directly onto the crankshaft or flywheel, removing the belt limitation and increasing available assist and regenerative capability, but the motor remains permanently coupled to the engine, so engine drag still reduces the effectiveness of energy recovery. The P2 architecture places the motor between the engine and transmission and introduces clutch-based decoupling, allowing the engine to be disconnected from the driveline. This makes electric launch possible, reduces friction losses during recuperation, and increases control flexibility relative to P0 and P1 [4].

The functions each architecture can support are another factor for its efficiency benefits. Mild hybrid systems improve fuel economy primarily through engine idle elimination, regenerative braking, electric torque assist, and engine load point shifting. Stop–start systems eliminate fuel consumption during stationary operation, which is particularly important in congested urban traffic. Regenerative braking recovers a portion of the vehicle's kinetic energy during deceleration and stores it for later reuse, instead of dissipating it entirely in the friction brakes. Torque assist reduces transient engine load during acceleration and can move the engine closer to more efficient operating regions. Load point shifting similarly exploits the electric motor to increase or reduce engine load so that the engine spends more time near higher-efficiency zones on the BSFC map. Among the three architectures, P2 offers the broadest benefits as engine decoupling also enables electric launch and limited electric-only propulsion at low speed, which are not available to the same extent in P0 and P1 systems [1], [4].

Comparative literature consistently indicates that these physical differences present measurable performance differences. In a GT-ISE-based comparison of a baseline conventional vehicle, a 12 V start–stop vehicle, and 48 V P0, P1 and P2 mild hybrid configurations based on a 2018 Ford Focus 2.0 L, Liu et al. reported average fuel economy improvements of 13.5% for P0, 15.5% for P1 and 18.5% for P2 relative to the baseline vehicle [4]. Under WLTP specifically, the same study reported gains of 18% for P0, 21% for P1 and 27% for P2, confirming that the performance ranking presents P2 as the optimal solution under a representative modern drive cycle [4]. The corresponding average CO_2 emissions were reduced from 179.4 g/km for the baseline vehicle to 157.9 g/km, 155.5 g/km and 151.8 g/km for P0, P1 and P2 respectively [4]. This evidence is particularly important because it isolates architecture as the main variable when keeping the base vehicle platform and driving conditions consistent.

Other comparative studies support the same general trend. Anselma et al. showed that parallel hybridisation can significantly reduce CO_2 emissions and improve drivability, with architectures that offer more flexible power flow and stronger electric participation generally outperforming simpler configurations under both standard and real-world driving [5]. Likewise, Luján et al. demonstrated that hybrid powertrains provide their greatest benefits in low-speed and urban operation, where fuel consumption reductions can be far greater than those seen on highway sections because regenerative braking, stop–start operation and electrically assisted launch are used more frequently [7]. Together, these studies show that the benefit of P2 is not an isolated result from one model, but part of a wider pattern in the hybrid vehicle literature which is that the more effectively the architecture supports recuperation, low-speed electric operation and engine decoupling, the greater the efficiency.

The importance of drive cycle selection follows directly from this point. Hybrid powertrains do not deliver a constant efficiency advantage across all operating conditions. Their benefits are maximised in transient, low-speed, stop–start environments where deceleration events are frequent and the engine would otherwise operate inefficiently. Experimental work on multiple generations of Toyota hybrid vehicles under urban conditions showed that more than 50% of the travelled distance could be completed in electric mode in most cases, regardless of initial battery charge, and that urban operation offered a highly suitable environment for hybrid propulsion because charging during regenerative braking and engine-assisted charging exceeded battery discharge over the route [8]. The same study showed strong interaction between discharge energy, regenerative energy recovery and operating mode share, reinforcing the importance of analysing the electrical energy path rather than only fuel consumption [8]. This is relevant to P0, P1 and P2 comparison because architecture determines how effectively braking energy can be recovered and reused.

Since hybrid benefits are highly cycle-dependent, the use of WLTP rather than NEDC is justified both technically and regulatorily. The NEDC became increasingly unrepresentative of real-world driving because it used outdated speed profiles, modest accelerations and unrealistically gentle operating conditions. The introduction of WLTP aimed to reduce the gap between laboratory values and real on-road behaviour by using longer test durations, higher average and maximum speeds, more dynamic acceleration and deceleration, reduced idle proportions, and vehicle-specific consideration of mass, aerodynamics and optional equipment [9], [10]. The WLTP cycle also divides operation into four phases, being low, medium, high and extra-high speed, representing urban, suburban, rural and motorway conditions respectively, which makes it better suited to evaluating how architecture-dependent hybrid benefits change across different driving conditions [10]. For this reason, WLTP Class 3 is a more suitable basis for comparing P0, P1 and P2 than NEDC.

The literature also supports simulation as an appropriate basis for such comparisons. Hybrid vehicles are inherently multi-domain systems involving mechanical drivetrain dynamics, engine operating maps, electric motor performance, battery behaviour and control. Bera emphasised that hybrid drivetrain modelling requires simultaneous treatment of rolling resistance, aerodynamic drag, inertial forces, engine efficiency, motor characteristics, inverter behaviour and battery constraints, and showed how these parameters interact in the estimation of vehicle energy demand and fuel consumption over a drive cycle [6]. Similarly, Niu and Abdullayev argued that model-based design in MATLAB/Simulink provides a practical way to represent the interactions among battery, motor-generator, power electronics, ICE and vehicle dynamics before physical testing, allowing control strategies and component behaviour to be evaluated in a common environment [11]. These modelling studies support the idea that architecture comparison under identical simulated conditions is a valid way to isolate the benefits of powertrain layout.

Within such simulations, the control strategy remains important because the energy management strategy determines how the hardware are used optimally. Control strategies vary significantly across P0–P4 architectures due to their differing physical configurations. This introduces a potential inconsistency in comparative studies, as the control strategy itself becomes a varying parameter rather than a controlled constant. Rule-based controllers remain common because they are simple and suitable for real-time implementation, but optimisation-based methods such as dynamic programming, ECMS and MPC generally achieve better fuel economy because they formalise the trade-off between fuel use, battery SOC and torque demand [3].

Overall, the literature establishes a foundation for this study. Hybridisation is a practical route to improved fuel economy and lower emissions in the near term, especially through 48 V mild hybrid systems [1]. Within parallel hybrids, P0, P1 and P2 form a logical progression in electrification capability, with P2 offering the greatest flexibility because of engine decoupling and electric launch [4]. However, the extent to which these theoretical advantages are seen in practice depends strongly on the control strategies for the architecture, which can significantly influence energy distribution and overall system performance. Therefore, a controlled comparison of P0, P1 and P2 architectures under identical drive cycle conditions is required to isolate the effect of architecture on system performance.

3) Methodology

The comparative study was conducted using the MATLAB/Simulink Powertrain Blockset, with P0, P1 and P2 architectures. These architectures were selected as they provide a controlled progression in hybridisation capability while maintaining comparable system boundaries, allowing the effect of motor placement on system performance to be isolated.

The P0 architecture utilises a belt-driven starter generator connected to the front of the ICE. This configuration is permanently coupled to the engine and does not allow electric-only operation. The P1 architecture places the electric motor directly on the crankshaft, maintaining a permanent mechanical connection to the ICE. The P2 architecture positions the electric motor between the ICE and transmission and incorporates a K0 decoupling clutch, enabling engine disengagement, electric-only operation and improved regenerative braking capability. The relative location of the electric motor and any disconnect clutches are shown in Figure 1:

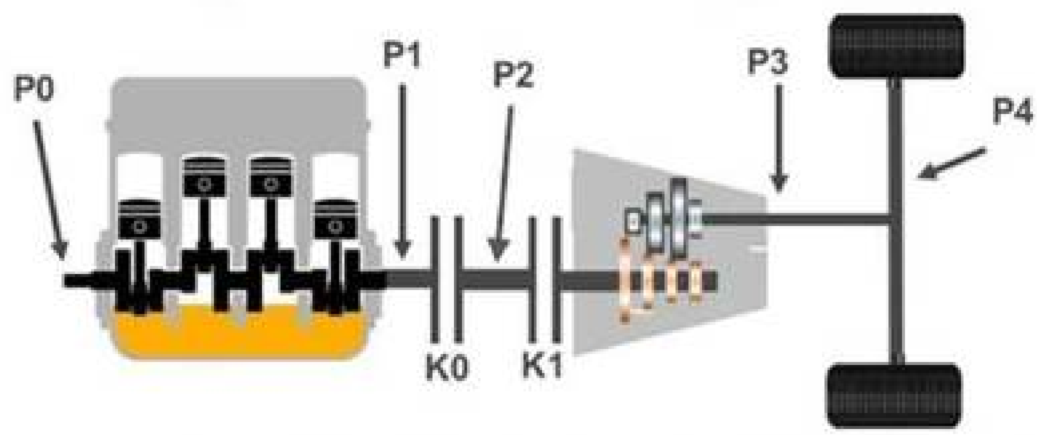

Figure 1: Schematic representation of P0–P4 hybrid architectures showing the positioning of the electric motor along the drivetrain and the presence of decoupling clutches (K0 & K1)

The WLTP Class 3 drive cycle was chosen because its aggressive accelerations and higher top speeds better represent modern, EU, real-world driving compared to older cycles like the NEDC. Class 3 was chosen over the Class 1 or 2 as the 1623kg vehicle mass puts it in the category of modern passenger cars. The drive cycle plot is shown in Figure 2:

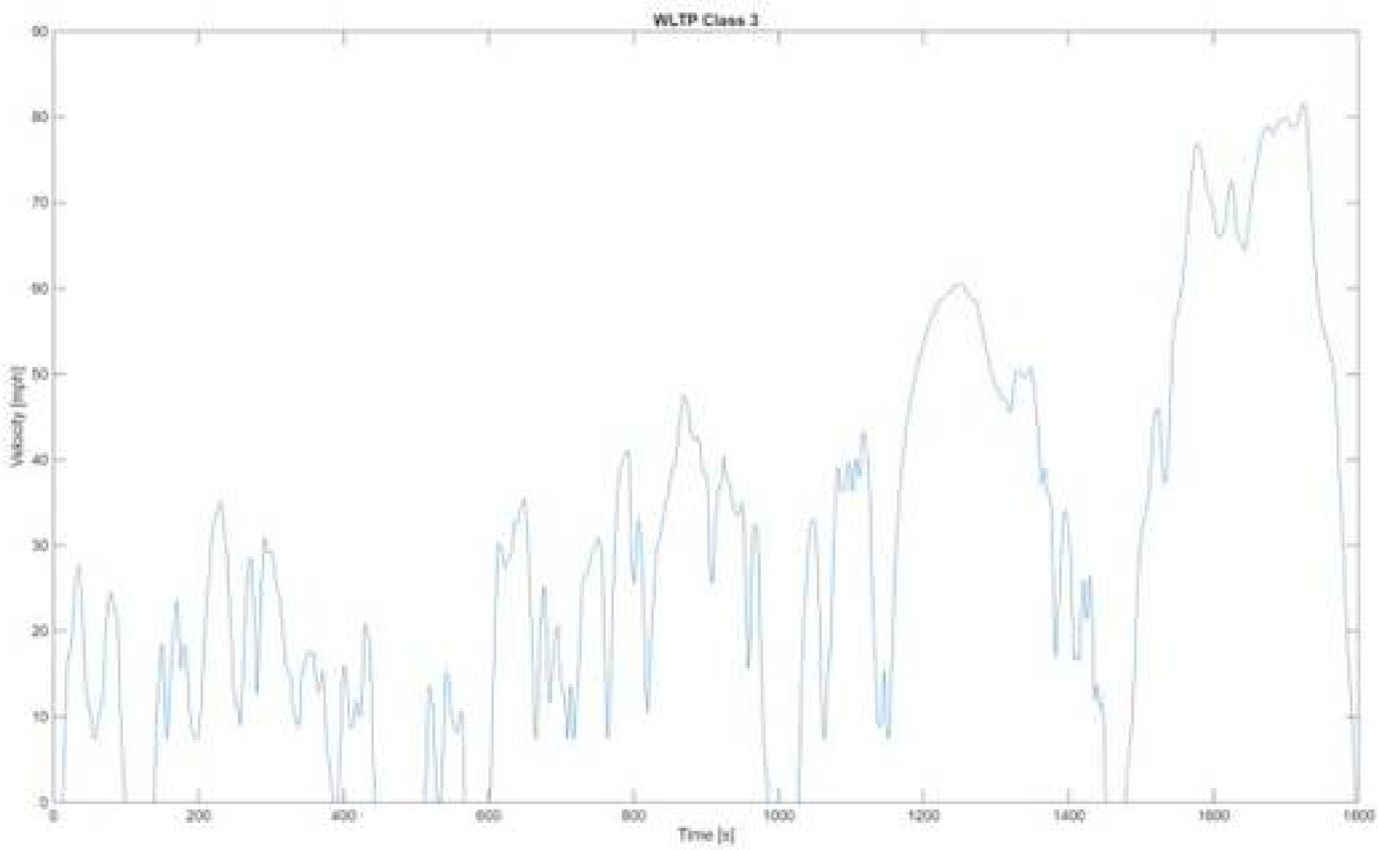

Figure 2: WLTP Class 3 drive cycle, showing vehicle velocity changes over the 1800s test duration.

To make sure the results accurately isolate the differences in the hybrid architectures, the parameters used in the models were extracted from the MATLAB dictionaries, where we can see all important parameters are identical across models, shown in Tables 1 and 2:

Table 1: Common model parameters applied across all simulation models to ensure a consistent comparison.

Parameter	Value
Vehicle Mass	1623 kg
Aerodynamic Drag Coefficient	0.23
Frontal Area	2.27 m²
Loaded Wheel Radius	0.327 m

Nominal Bus Voltage	400 V
Gross Battery Capacity	2.88 Ah
Target Battery SOC	60%

Table 2: Architecture-specific model parameters for P0, P1 and P2 configurations.

Parameter	P0	P1	P2
Primary Motor Gear Ratio	3:1	1:1 (Direct)	2.636:1
Peak Motor Torque (Nm)	66.67	200.0	225.0
Pure EV Capability (Clutch)	No	No	Yes

Accurate fuel economy evaluation in HEVs requires accounting for changes in battery SOC. A decrease in SOC leads to artificially high fuel economy, while an increase results in artificially low values. To eliminate this bias, all simulations were conducted under charge-sustaining conditions, where initial and final SOC are equal.

Energy management was implemented using an Equivalent Consumption Minimisation Strategy (ECMS), which determines the optimal power split between the ICE and electric motor. The ECMS weighting factor was calibrated for each architecture using a MATLAB script based on the Secant method. Starting from an initial SOC of 60%, ECMS_s was iteratively adjusted until the final SOC converged within ±0.1%, ensuring zero net battery energy change over the drive cycle. It should be noted that although ECMS ensures charge-sustaining operation, the calibration may influence how aggressively energy is utilised, meaning that control strategy can affect the performance of each architecture. The calibrated values are shown in Table 3:

Table 3: Comparison of default and calibrated ECMS weighting factors for P0, P1 and P2 architectures.

Model	Standard Dictionary ECMS Value	Calibrated MATLAB ECMS Value
P0	2.2282	2.0195
P1	4.4644	3.4901
P2	4.3080	3.8608

Signals from the Performance and Emissions scopes were logged directly to the MATLAB base workspace during each simulation in a Structure With Time format. A MATLAB script was used to extract the time-series data, and export it as CSV files for analysis in Excel. The outputs selected for analysis included target and actual vehicle velocity; engine and motor speed; engine and motor torque; battery current and SOC; cumulative fuel economy; and cumulative emissions of hydrocarbons (HC), carbon monoxide (CO), nitrogen oxides (NOx), and carbon dioxide (CO_2).

4) Results & Discussion

The velocity tracking error was calculated which is defined as actual vehicle speed minus target speed, for P0, P1 and P2 configurations over the WLTP Class 3 cycle, shown in Figure 3 below:

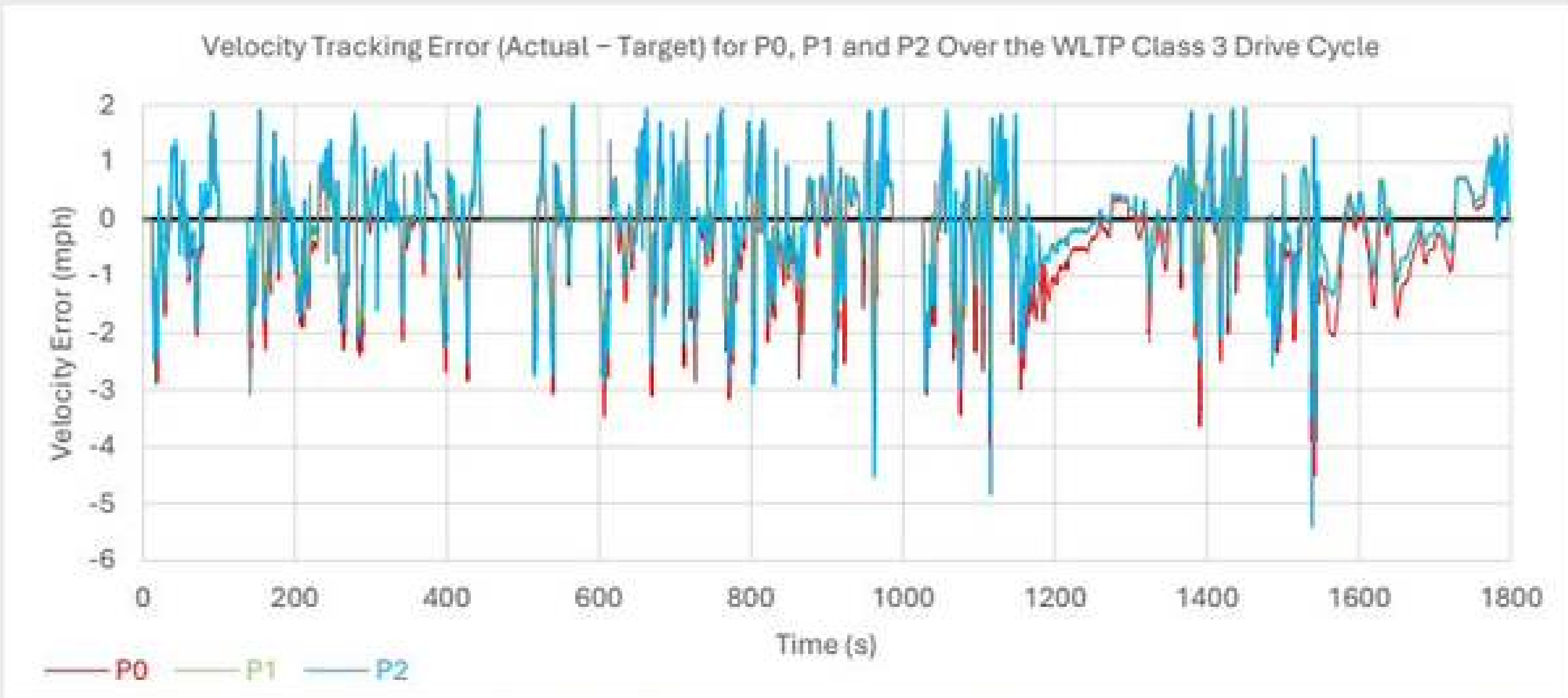

Figure 3: Velocity tracking error for P0, P1 and P2 architectures showing the deviation from the target velocity profile.

Across all three architectures, the tracking error remains relatively small for most of the cycle, indicating generally good tracking to the drive cycle. However, clear differences in performance are observed. P1 maintains the smallest deviations demonstrating the best overall tracking. P0 exhibits slightly larger negative deviations, indicating a tendency to undershoot the target speed. P2 shows the largest and most irregular errors, including high negative spikes. The poorer tracking observed in P2 suggests less stable torque delivery under transient conditions, which can cause negative downstream effects on fuel consumption and emissions.

The powertrain behaviour of P0, P1, and P2 was subsequently analysed by examining engine and motor speeds alongside their respective torque. To capture both low-speed and higher-speed operating conditions within the drive cycle, two time intervals are considered, being 130s to 400s and 1500s to 1800s. These windows enable a focused comparison of how each architecture distributes propulsion effort between the ICE and electric motor under different driving conditions. The engine speed compared between P0, P1 and P2 across the low speed section are shown in Figure 4:

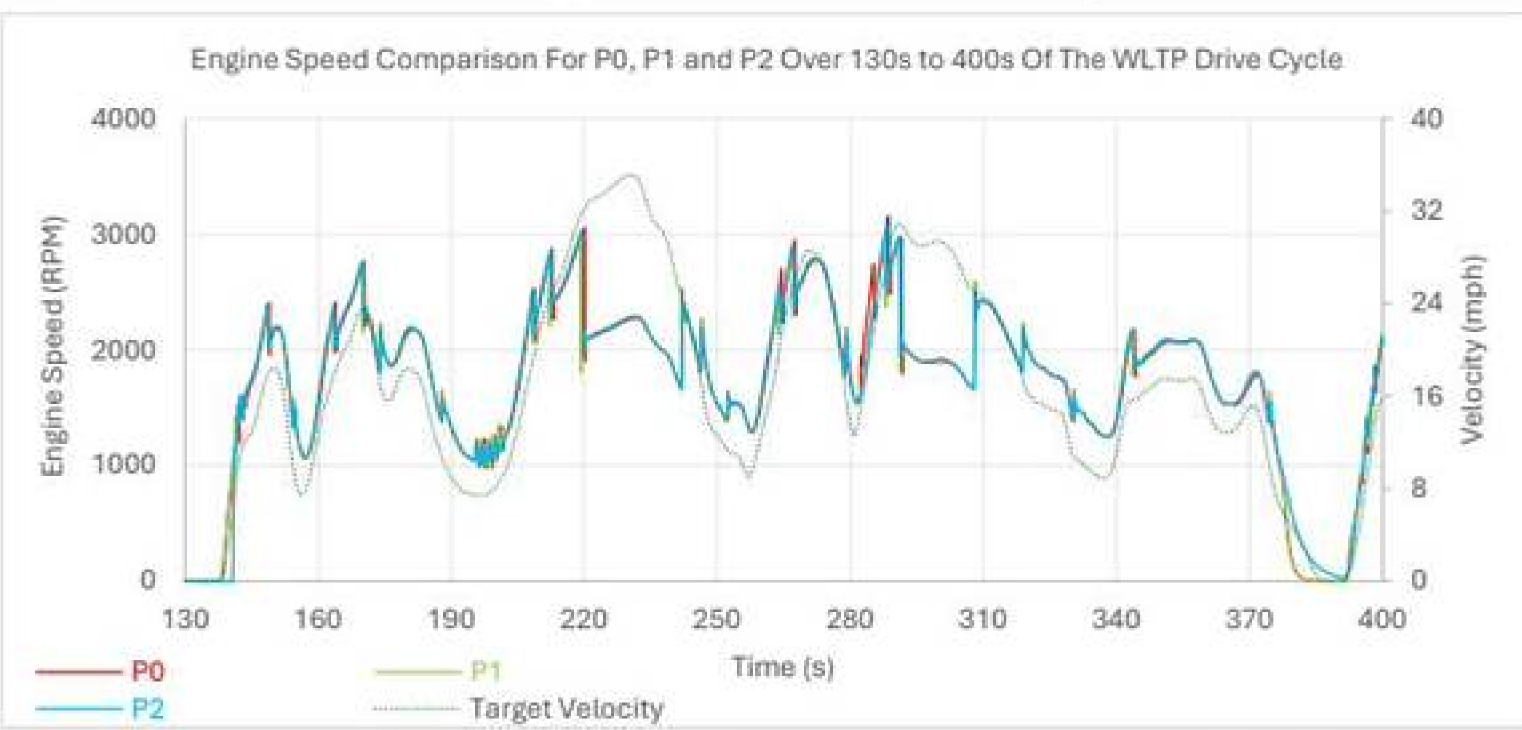

Figure 4: Engine speed response of the P0, P1 and P2 architectures over a low speed section of the drive cycle.

The three traces are closely aligned throughout most of the cycle, indicating that the engine speed is similar across all three configurations. In all cases, engine speed rises and falls in the same manner, with the main peaks occurring during the more demanding portions of the cycle. However, the differences in the powertrain layout are more clearly visible in the motor speed comparison, shown in Figure 5:

Figure 5: Motor speed response of the P0, P1 and P2 architectures over a low speed section of the drive cycle.

Unlike the engine speed behaviour, the motor speed traces do not overlap closely. The P0 architecture operates at higher motor speed than the other two configurations where the P1 and P2 traces remain much lower and are closely grouped together throughout most of the cycle. This shows how differently the motors in each architectures are designed to function. This behaviour is primarily due to the differing motor gear ratios, where the P0 configuration (3:1 ratio) results in significantly higher motor rotational speeds relative to the engine, whereas P1 (1:1) and P2 (2.636:1) operate at lower speeds with greater torque capability.

The activity of the engine and motor Is seen clearer in the torque responses for P0, P1 and P2 over the 130–400 s interval as shown in Figures 6–8:

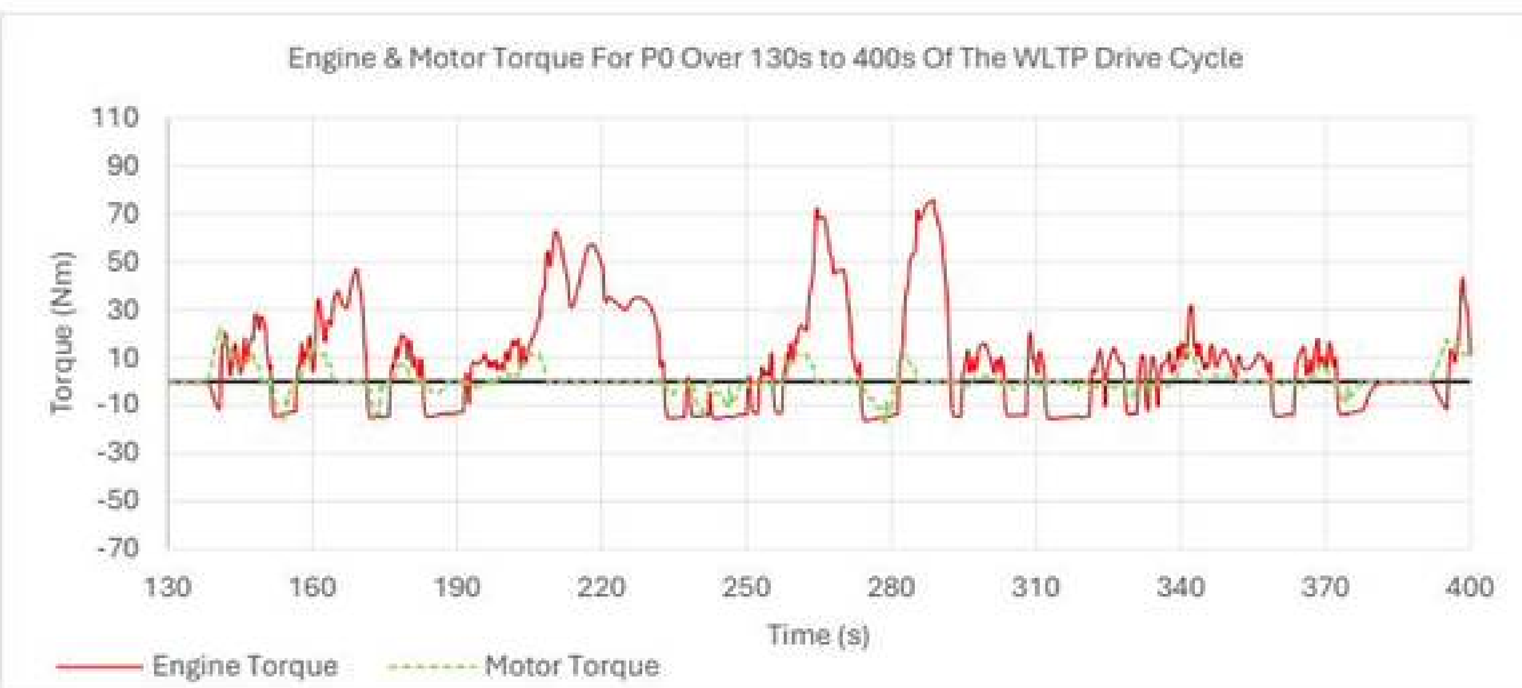

Figure 6: Powertrain torque response of the P0 architecture over a low speed section of the drive cycle.

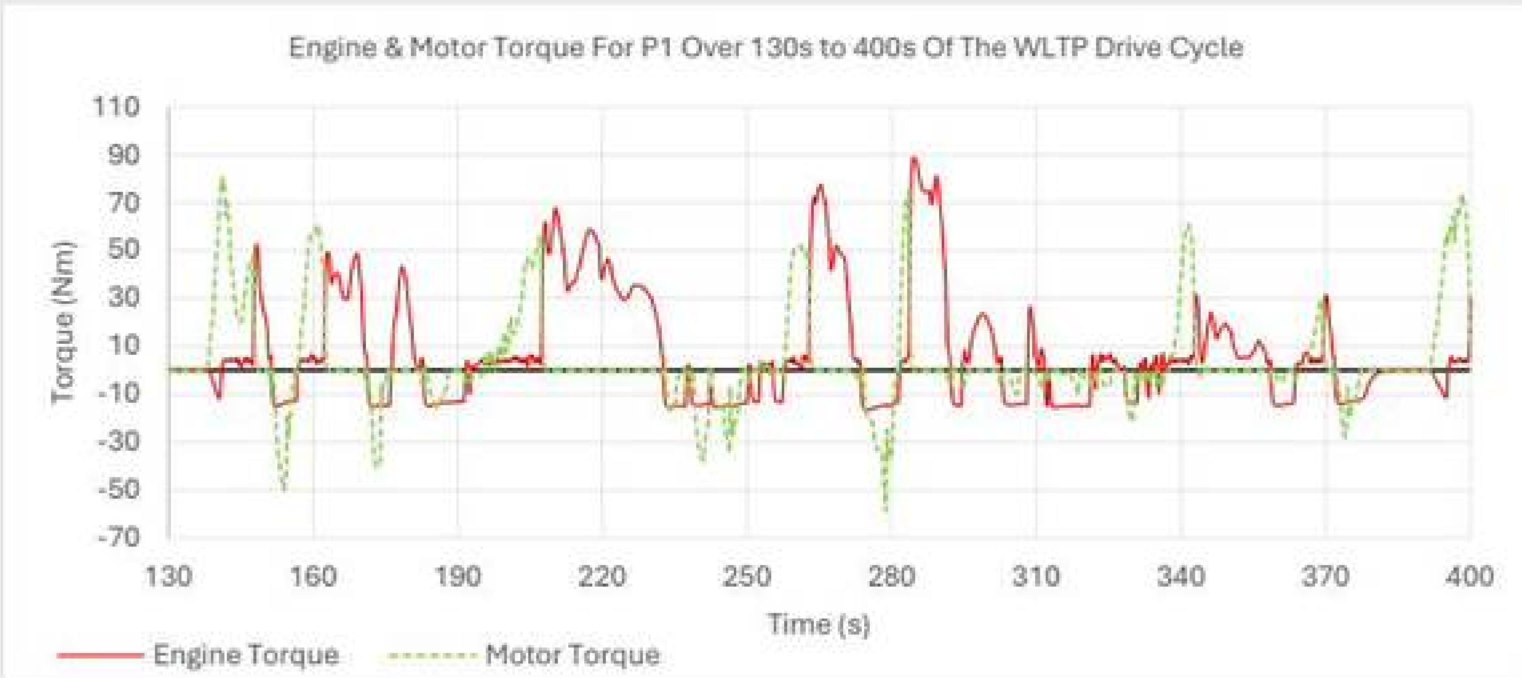

Figure 7: Powertrain torque response of the P1 architecture over a low speed section of the drive cycle.

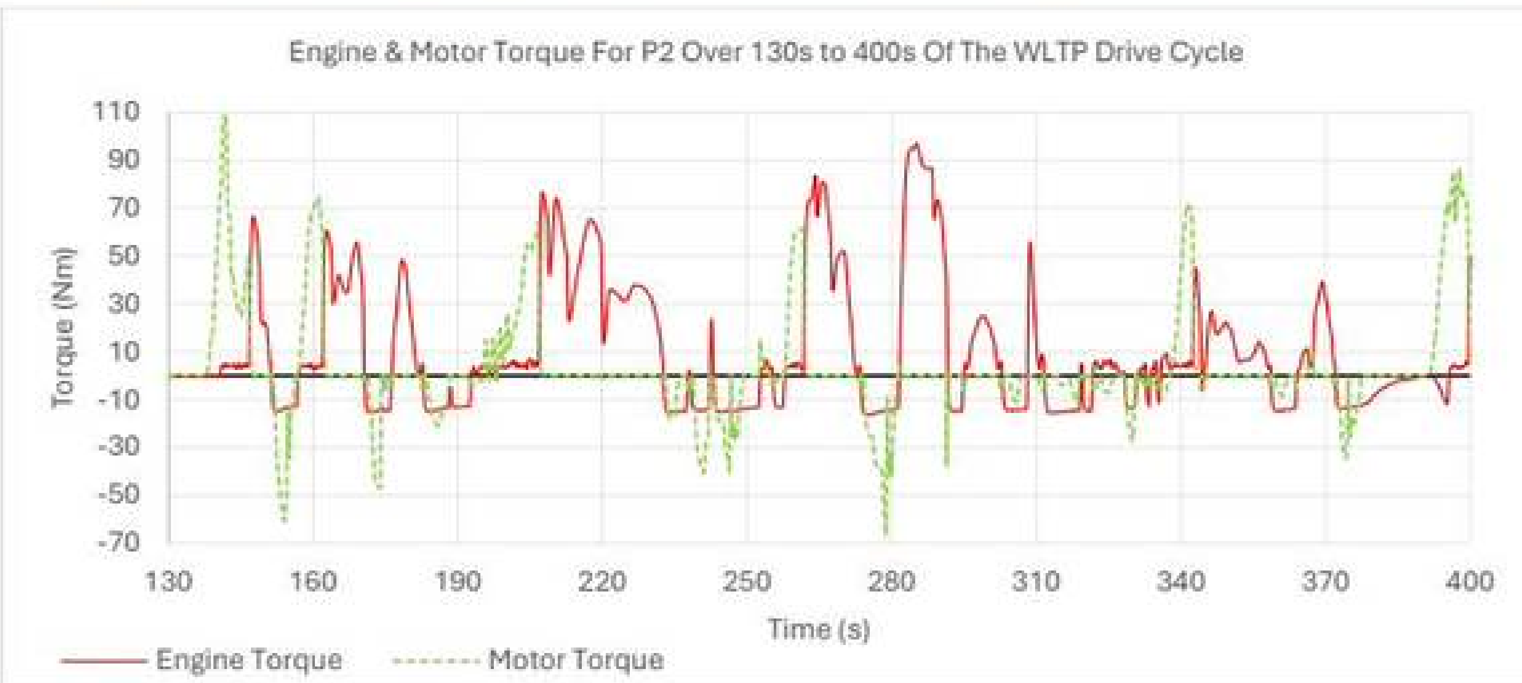

Figure 8: Powertrain torque response of the P2 architecture over a low speed section of the drive cycle.

Across all three configurations, the ICE remains the primary source of positive torque. However, the level of electric motor contribution increases significantly with hybridisation. The P0 configuration uses the engine mostly with the motor contributing near-zero assist and regeneration. The motor in the P1 configuration becomes more active, showing larger bidirectional torque spikes and contributing torque during both assist and recovery phases. This trend is higher in the P2 configuration where the motor delivers the largest positive and negative torque events, in some regions approaching the magnitude of the engine contribution which is enabled by the disconnect clutch, which allows the electric motor to operate independently of engine inertia. Overall, while engine behaviour remains broadly consistent across the architectures, the key differences lies in the progressively stronger and more dynamic contribution of the electric motor from P0 to P2, highlighting the increased hybrid functionality and control flexibility with higher levels of integration.

The torque responses for P0, P1 and P2 can also be seen over the higher speed 1500s to 1800s interval as shown in Figures 9-11. Compared to the earlier 130s to 400s window, this phase is more engine-dominated across all configurations:

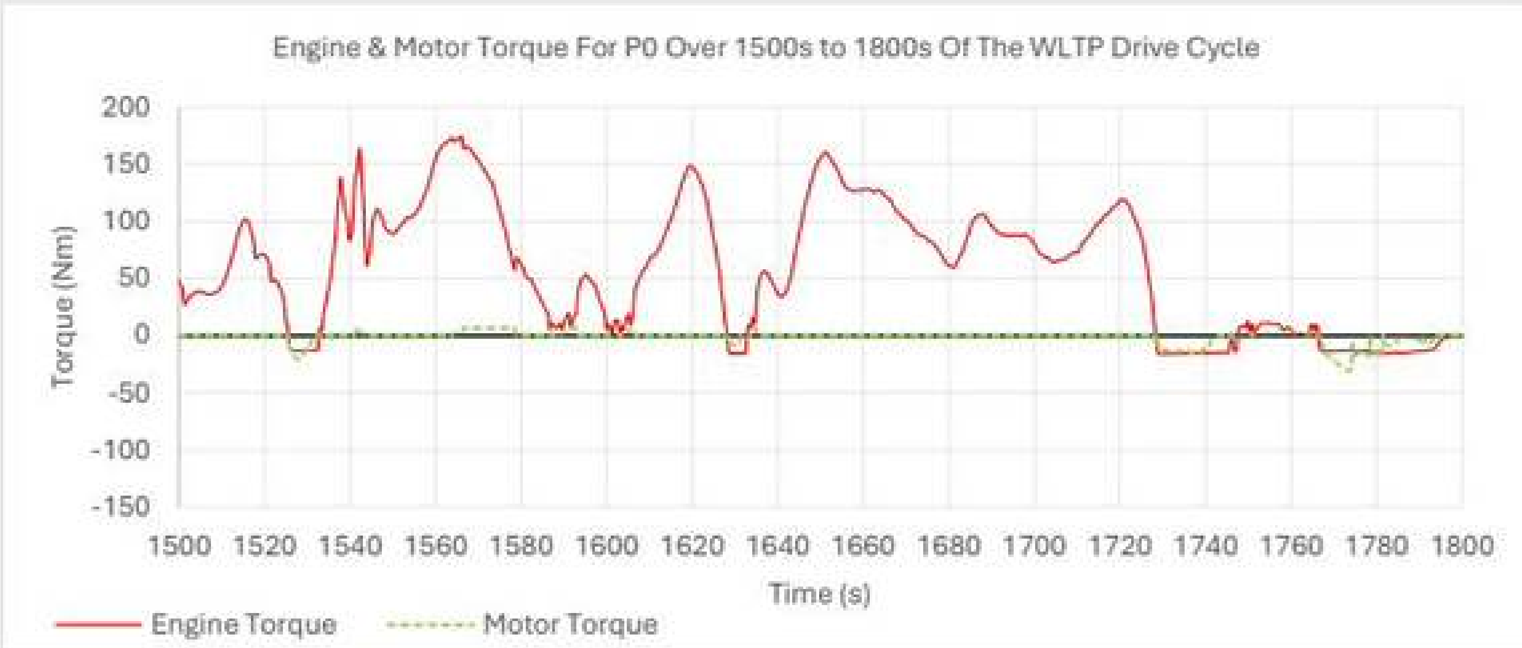

Figure 9: Powertrain torque response of the P0 architecture over a high speed section of the drive cycle.

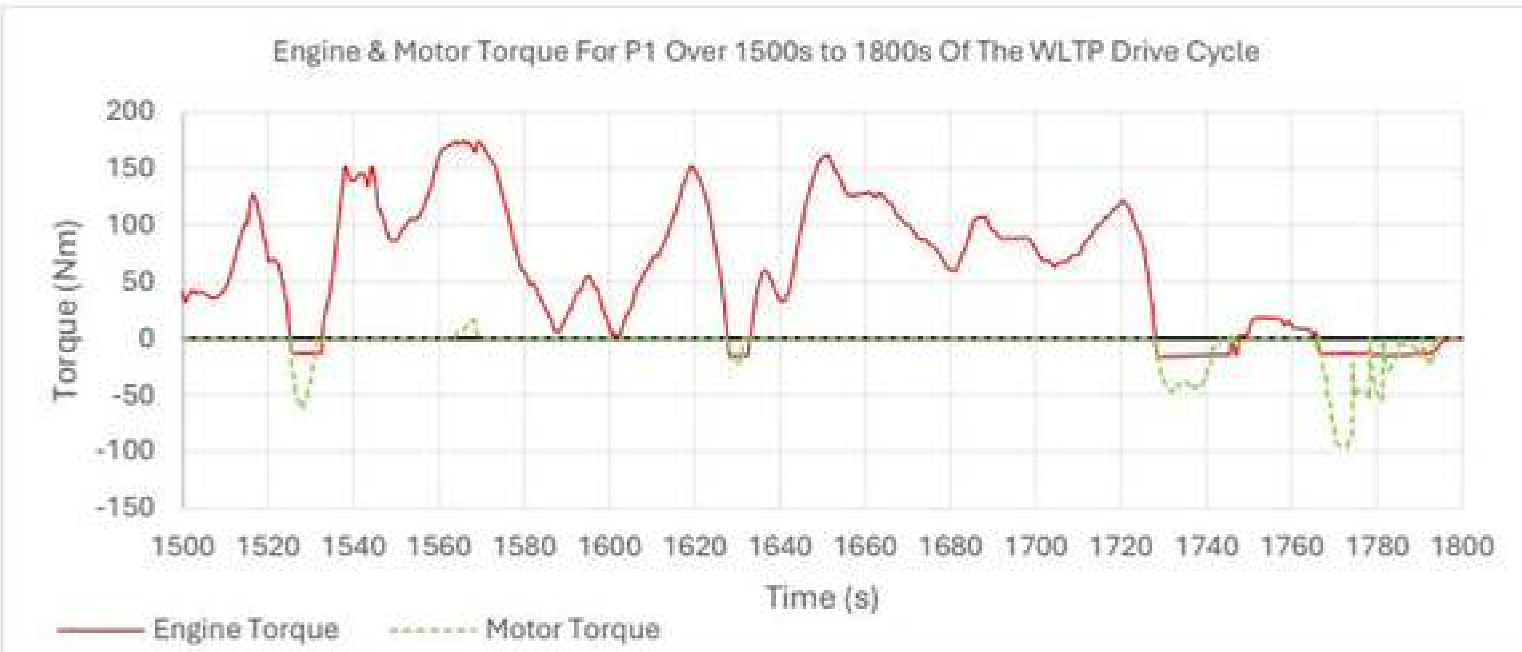

Figure 10: Powertrain torque response of the P1 architecture over a high speed section of the drive cycle.

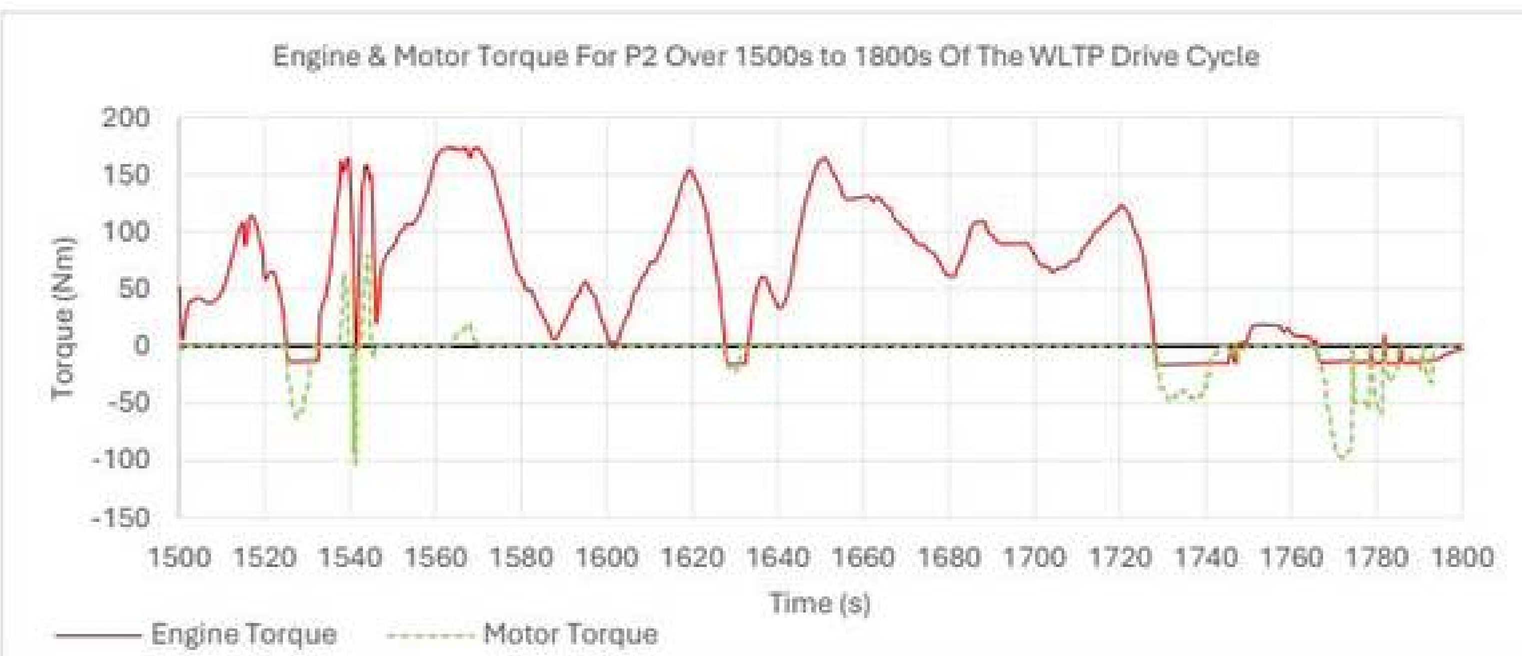

Figure 11: Powertrain torque response of the P2 architecture over a high speed section of the drive cycle.

Taken together, the torque results across both intervals show that while engine behaviour remains broadly consistent between P0, P1 and P2, the key distinguishing factor is the level of electric motor contribution. In the 130s to 400s interval, differences are more visible, with P0 showing limited motor torque and P1 and especially P2 showing increasingly higher bidirectional motor-torque activity, indicating greater electric assist during operation. In contrast, the 1500s to 1800s interval is more engine-dominated across all architectures. However, P0 maintains the lowest motor contribution, P1 shows moderate regenerative contribution, and P2 continues to show the largest transient motor-torque spikes. Overall, increasing hybrid integration primarily increases the dynamic contribution of the electric motor rather than significantly altering engine behaviour, with this effect most visible during transient operating conditions. This increased engine dominance is consistent with higher-speed operation, where sustained power demand reduces opportunities for regenerative braking and limits the relative contribution of the electric motor.

The battery state of charge over the drive cycle for the P0, P1 and P2 architectures is shown below in Figure 12:

Figure 12: Battery SOC of the P0, P1 and P2 architectures over the full WLTP Class 3 drive cycle.

All three configurations begin at the MATLAB initial value of 60% SOC and exhibit an initial discharge phase, reaching minimum values at around 800s before recovering during the second half of the cycle. Clear differences are visible between the architectures. The P0 configuration maintains the highest SOC throughout most of the cycle, while P1 reaches the lowest SOC and appears to use the battery most aggressively. P2 remains between the two. By the end of the cycle, all three traces return to approximately their starting SOC, indicating broadly charge-sustaining operation over the complete drive cycle which was optimised through the ECMS factor.

The fuel-economy response comparison of the P0, P1 and P2 architectures over the full WLTP Class 3 drive cycle is seen below in Figure 13:

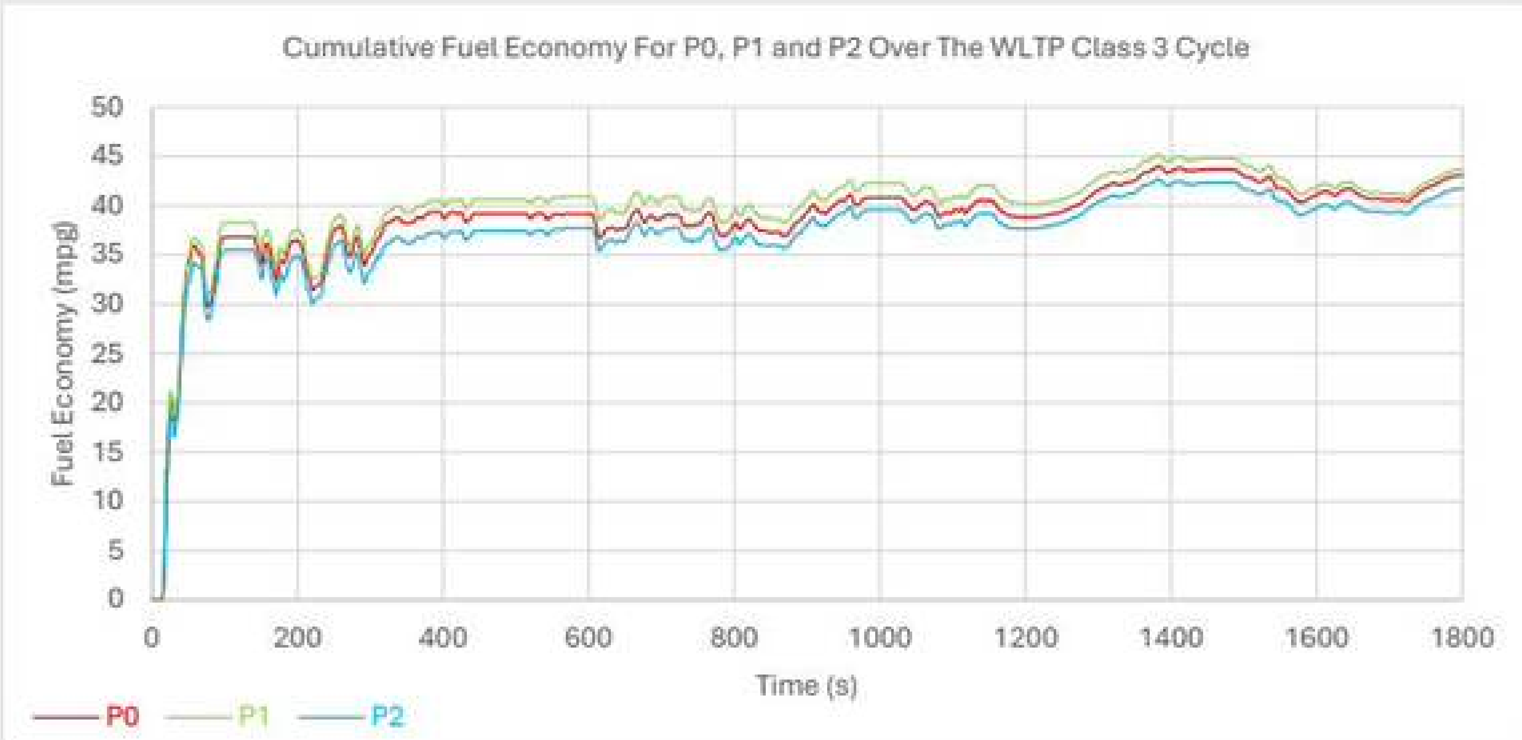

Figure 13: Fuel economy of the P0, P1 and P2 architectures over the full WLTP Class 3 drive cycle.

Following the initial transient region, a consistent ranking is observed for most of the cycle. The P1 configuration achieves the highest fuel-economy values, P0 remains slightly lower, and P2 gives the lowest overall performance. Although the differences are small, the ordering is constant throughout most of the cycle and remains visible at the end of the simulation. This indicates that, under the WLTP Class 3 operating conditions considered here, P1 provides the best fuel-economy performance, while P2 is the least efficient of the three architectures.

Cumulative emissions of HC, CO, NOx and CO2 were evaluated to compare the performance of the P0, P1 and P2 hybrid architectures over the WLTP Class 3 drive cycle. The HC emissions are split across each half of the drive cycle shown in Figure 14 and 15 below:

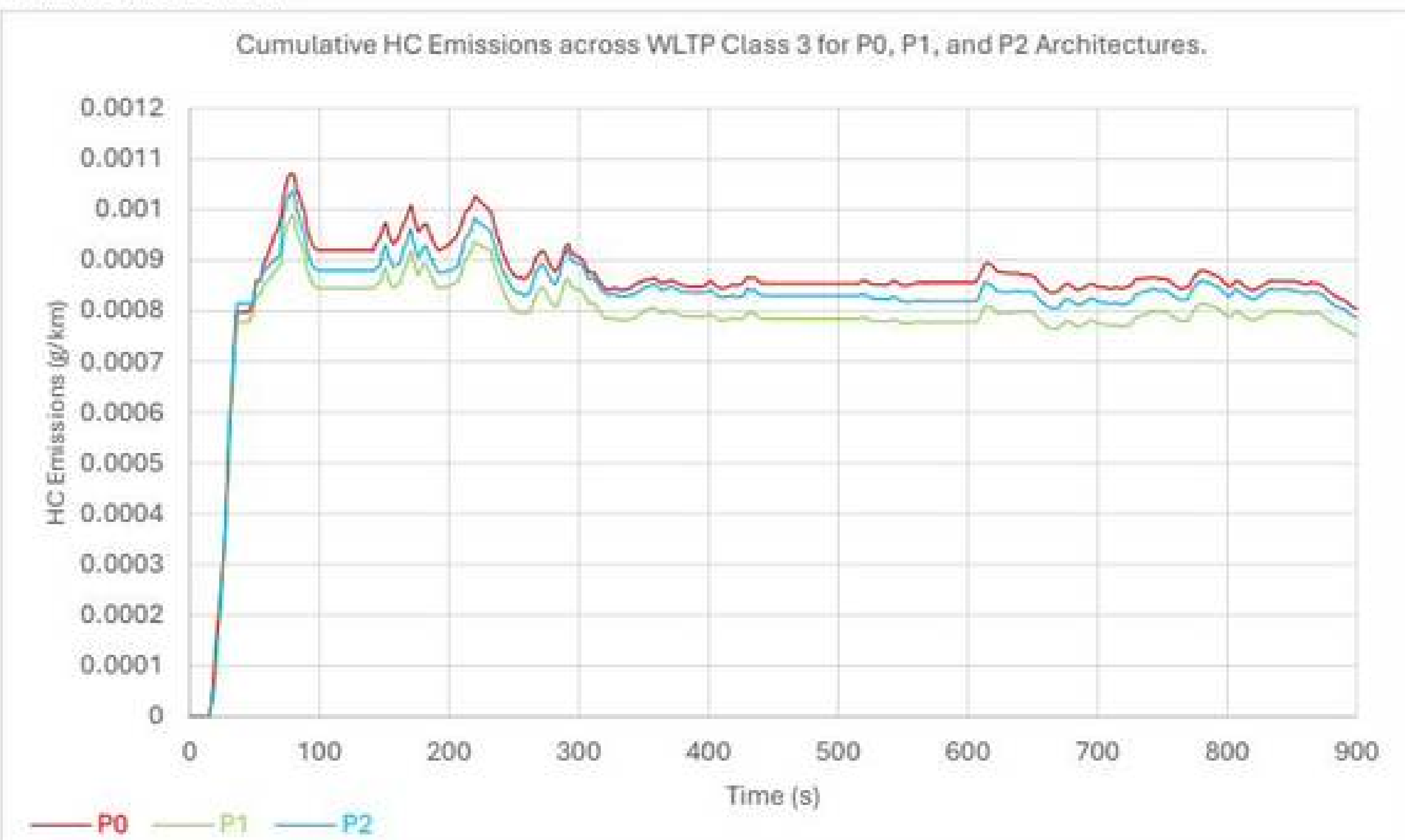

Figure 14: Cumulative HC emissions of the P0, P1 and P2 architectures over the 1st Half of the drive cycle.

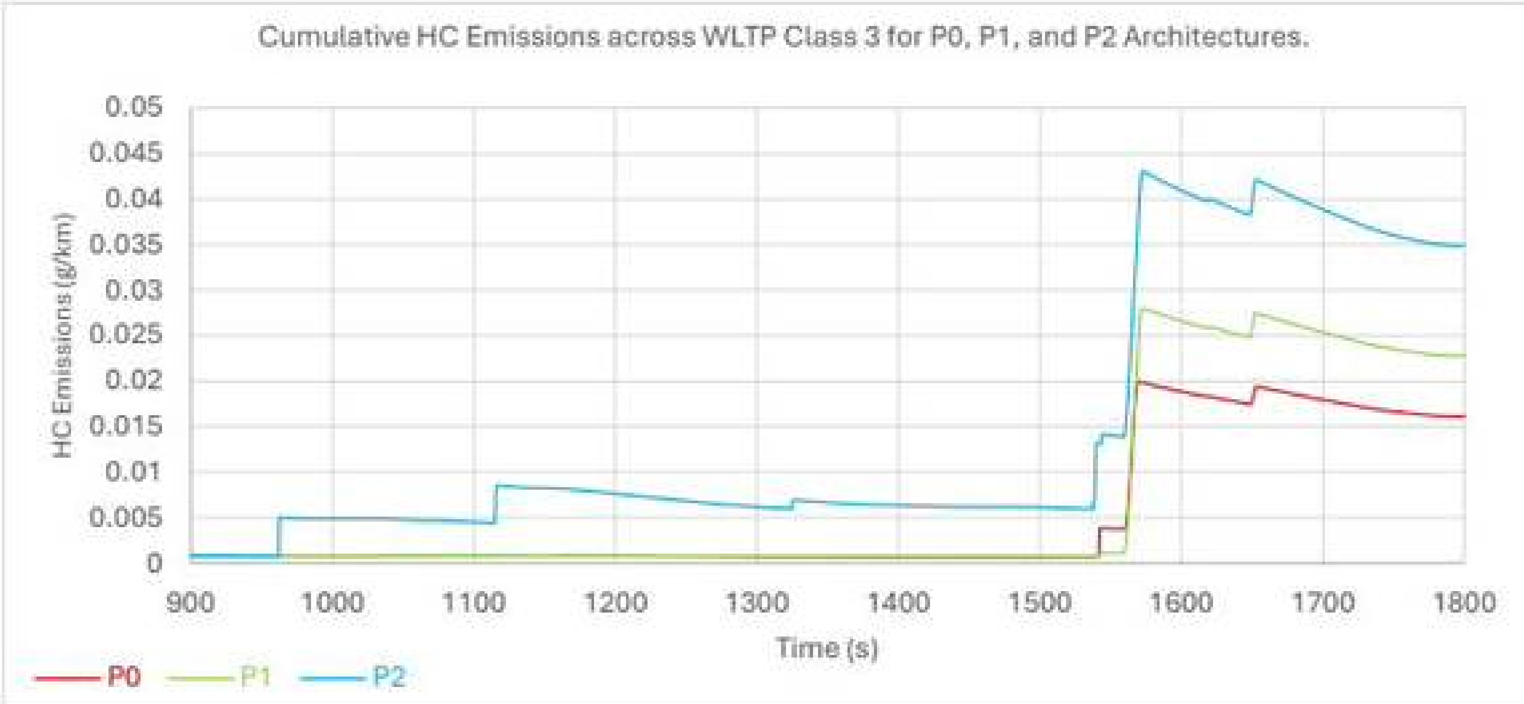

Figure 15: Cumulative HC emissions of the P0, P1 and P2 architectures over the 2nd Half of the drive cycle.

The CO emissions are also split across each half of the drive cycle shown in Figure 16 and 17 below:

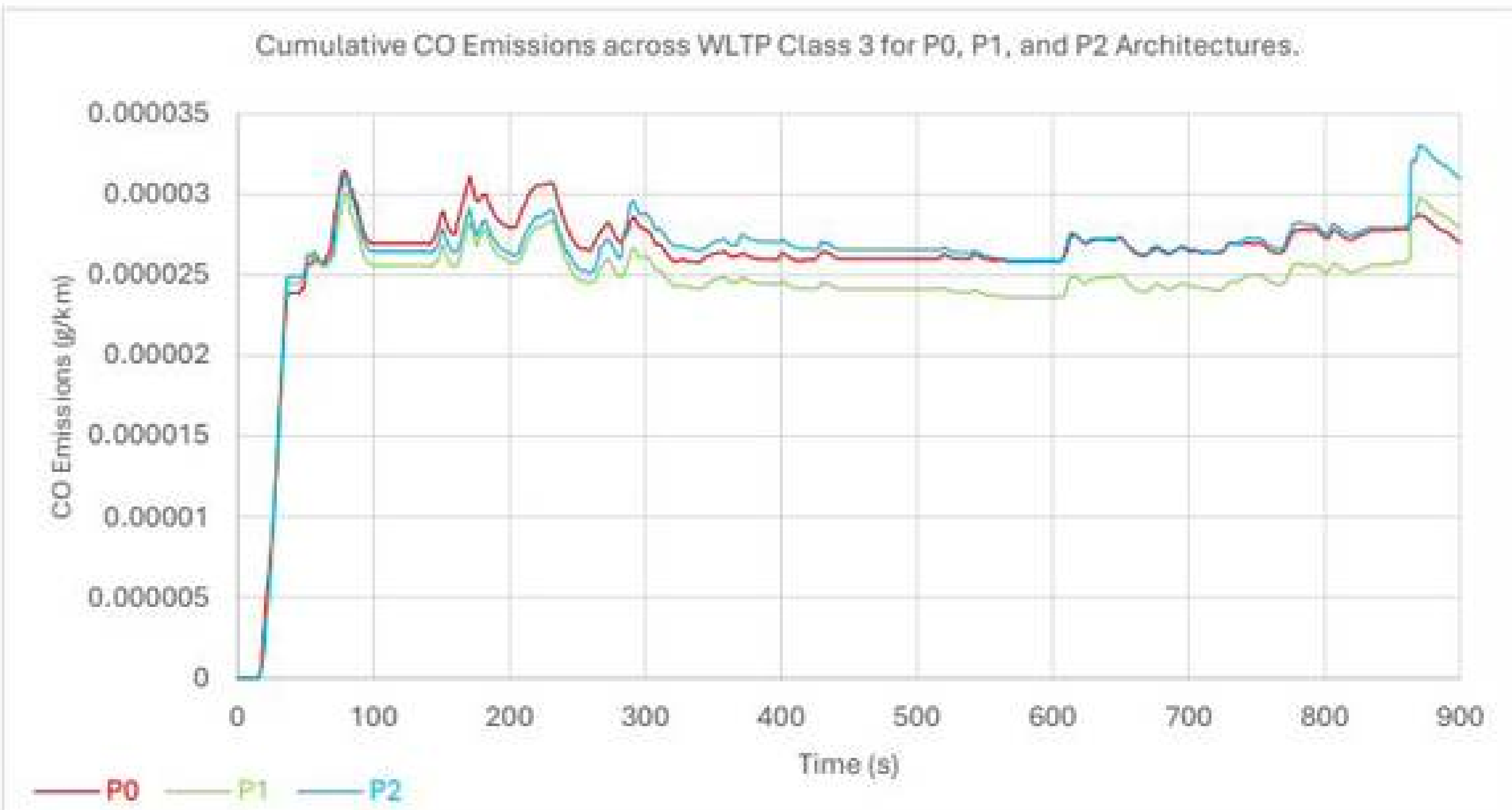

Figure 16: Cumulative CO emissions of the P0, P1 and P2 architectures over the 1st Half of the drive cycle.

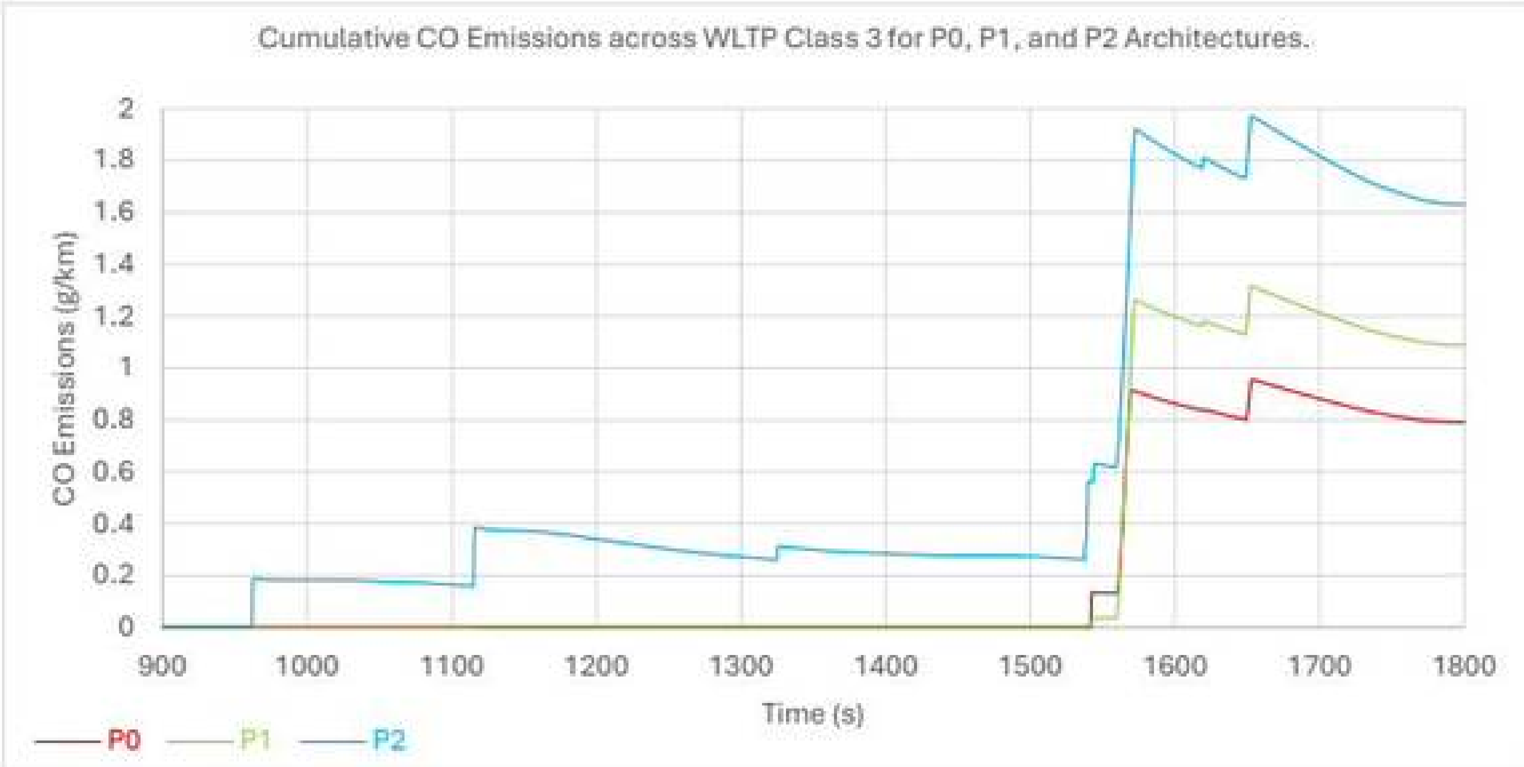

Figure 17: Cumulative CO emissions of the P0, P1 and P2 architectures over the 2nd Half of the drive cycle.

The cumulative NOx and CO2 emissions can be seen in Figure 18 and 19 below:

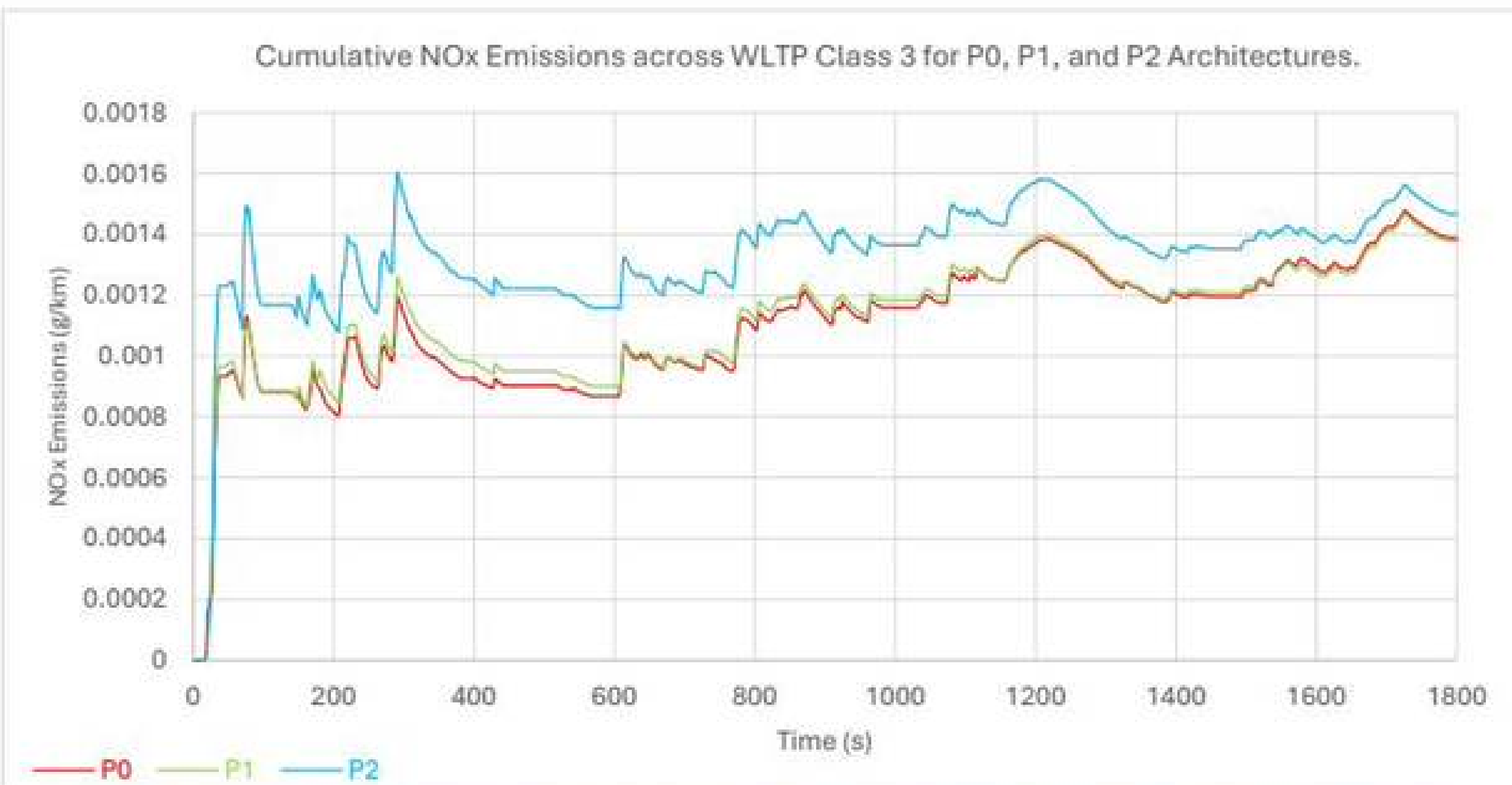

Figure 18: Cumulative NOx emissions of the P0, P1 and P2 architectures over the full WLTP Class 3 drive cycle.

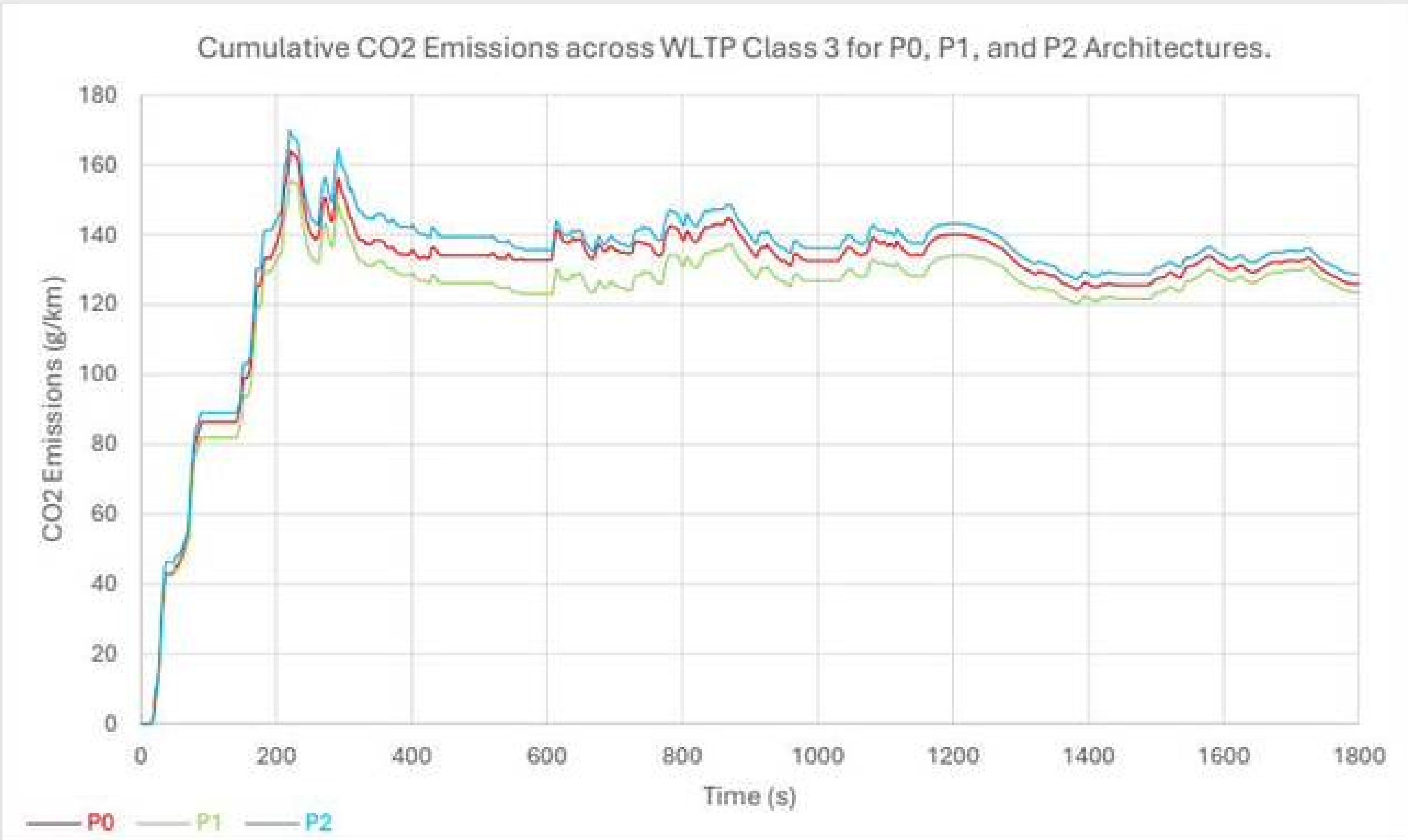

Figure 19: Cumulative CO2 emissions of the P0, P1 and P2 architectures over the full WLTP Class 3 drive cycle.

The cumulative emissions results show that both the magnitude and timing of pollutant formation are strongly influenced by hybrid architecture. For HC and CO, two distinct behaviours are observed across the WLTP cycle. During the first 900s, all three architectures follow a similar trend with small separation after an initial rise, indicating limited difference under lower-demand conditions. However, the second half of the cycle shows a huge spike around 1600s, which begins a divergence between configurations. In this region, P2 accumulates significantly higher HC and CO emissions than P0 and P1, while P0 and P1 alternate as the lower emitters depending on the pollutant. This demonstrates that late-cycle transients are the primary source of architecture-dependent variation for HC and CO.

In contrast, NOx and CO_2 exhibit more consistent behaviour over the full cycle. NOx emissions increase in a step-like manner with a ranking of P2 highest, P1 slightly above P0, and P0 lowest throughout. Similarly, CO_2 shows a clear and persistent ordering, with P2 producing the highest emissions, P0 intermediate, and P1 the lowest. While the absolute differences in CO_2 are smaller than for other pollutants, the consistent separation indicates that P1 delivers the best overall CO_2 performance, whereas P2 performs worst.

Overall, the results indicate that P2 exhibits the weakest emissions performance across most metrics whereas P1 provides the most balanced outcome, achieving the lowest CO_2 emissions and generally low CO, while P0 performs best in NOx and shows improved HC behaviour in the later stages of the cycle. These findings show that hybrid architecture influences not only total emissions but also the timing and severity of emissions release.

5) Conclusions

This study compared the performance of P0, P1 and P2 HEV architectures under identical WLTP Class 3 conditions using MATLAB/Simulink. The results confirm that hybrid architecture strongly influences transient behaviour, energy utilisation, fuel economy and emissions, consistent with literature linking motor placement to system performance. All architectures achieved acceptable velocity tracking, with P1 showing the smallest deviations and P2 the largest, indicating less stable transient response. While engine-speed behaviour remained similar, key differences arose in motor operation. In the 130s to 400s interval, P0 was engine-dominant with high motor speed and limited torque, whereas P1 and especially P2 exhibited stronger bidirectional motor torque. At higher speeds, all architectures became more engine-led, although P2 maintained the most dynamic motor activity due to its decoupling capability. This behaviour is directly linked to the mechanical configuration, where increased hybrid integration enables greater control flexibility but also introduces sensitivity to control calibration.

Battery behaviour further highlighted these differences. All models operated in charge-sustaining mode, returning to the initial 60% SOC. However, P1 and P2 showed deeper discharge, indicating greater electrical energy utilisation. Despite its higher flexibility, P2 delivered the lowest fuel economy, while P1 achieved the highest. This suggests that effective balance between engine operation and electric assistance, rather than maximum electrification, influences efficiency under the given control strategy. The higher ECMS weighting required for P2 further constrained electrical energy usage, limiting its ability to exploit its full hybrid capability.

Emissions results reinforced this trend. HC and CO diverged significantly during late-cycle transients, with P2 producing the highest values, while NOx and CO2 showed consistent rankings across the cycle. These late-cycle spikes correspond to high-load operating regions of the WLTP cycle, where rapid torque demand can drive the engine into less efficient combustion regimes. This highlights the strong coupling between transient control behaviour and emission formation.

Overall, although P2 offers the greatest theoretical capability, its poorer performance in this study is attributed to control limitations, including less stable torque delivery and constrained energy management under ECMS calibration. This highlights that increased hybridisation does not inherently guarantee improved performance, and that architecture effectiveness is strongly dependent on control strategy implementation. Under the conditions considered, P1 provides the most effective compromise between performance, efficiency and emissions, while P0 remains a simpler, engine-dominant solution. In contrast, P2 demonstrates that without optimised control, additional hardware capability can lead to diminished real-world benefits.

These findings are subject to the limitations of simulation-based analysis, including model reliability and control calibration. Results are also specific to the WLTP Class 3 cycle and may vary with alternative drive cycles, component sizing or control strategies. Future work should investigate optimised control approaches, particularly for higher levels of hybridisation, and validate these trends using higher-fidelity or experimental data.

6) References

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